

Capstone Project - HealthTrack-MAS: Multi-Agent System for Medical Record Aggregation

The idea:

HealthTrack-MAS — a multi-agent system where the user uploads all their medical PDFs from different clinics and hospitals and multiple specialized agents automatically collaborate to extract the data, build a health timeline across years, detect conflicts and gaps between reports, and generate a final patient-friendly health report. Chose this because it solves a real problem nobody does manually — most people have medical records scattered across 10 different places with no way to see the full picture.

The problem:

People often have medical records from different hospitals and doctors spread across many years. My system helps collect and organize those records, finds missing information, creates a timeline of their health history, and explains everything in simple language while answering questions about their medical records.

How the system works :

The user can upload medical pdf files or describe their medical history in text. The system then uses several AI agents that each have a specific job. One agent finds which medical records exist and which ones might be missing. Another agent reads the documents and extracts important medical information. Another agent organizes everything into a timeline. Another agent checks for contradictions or missing follow ups. And another agent writes a simple health summary. At the end the user can ask questions and the system answers using all the collected information.

Why this system is useful and different

Most existing tools only read one document at a time and explain it. My system is different because it works with many documents from different sources and tries to build a complete health history. It also does not just explain it, it finds gaps in the medical record and suggests what might be missing. It also organizes everything over time and helps the user understand patterns in their health in a simple way.